



Malla Reddy Engineering College for Women

UGC Autonomous Institution

Accredited by NBA & NAAC with A+ Grade

Approved by AICTE, Affiliated to JNTUH, ISO 9001:2015 Certified Institution

Maisammaguda, Dhulapally, Secunderabad-500100

Title of the Idea: LAND REGISTRY MANAGEMENT SYSTEM USING BLOCKCHAIN

TECHNOLOGY

Branch & Year: CSIT & IIIrd year

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PROBLEM STATEMENT :

Unlike developed countries, the property ownership details in developing countries isn't only inefficient, but is also error-prone and can be manipulated. Also several malpractices still exist in the system just due to lack of a transparent and immutable system. Therefore, we propose a Blockchain based solution to curb the flaws of the existing system.



KEY POINTS

- Decentralisation
- Smart Contracts
- Cryptography
- Transparent Ownership
- Secure data storage
- Faster Transactions



EXISTING INPUTS

EXISTING SYSTEM:

Feature	Description
Data Storage	Centralized, paper-based or siloed databases
Verification	Manual, slow, and error-prone
Fraud Risk	High – forgery, double-selling, tampering
Transparency	Low – limited access to records
Efficiency	Low – bureaucratic delays and



Benefit	Description
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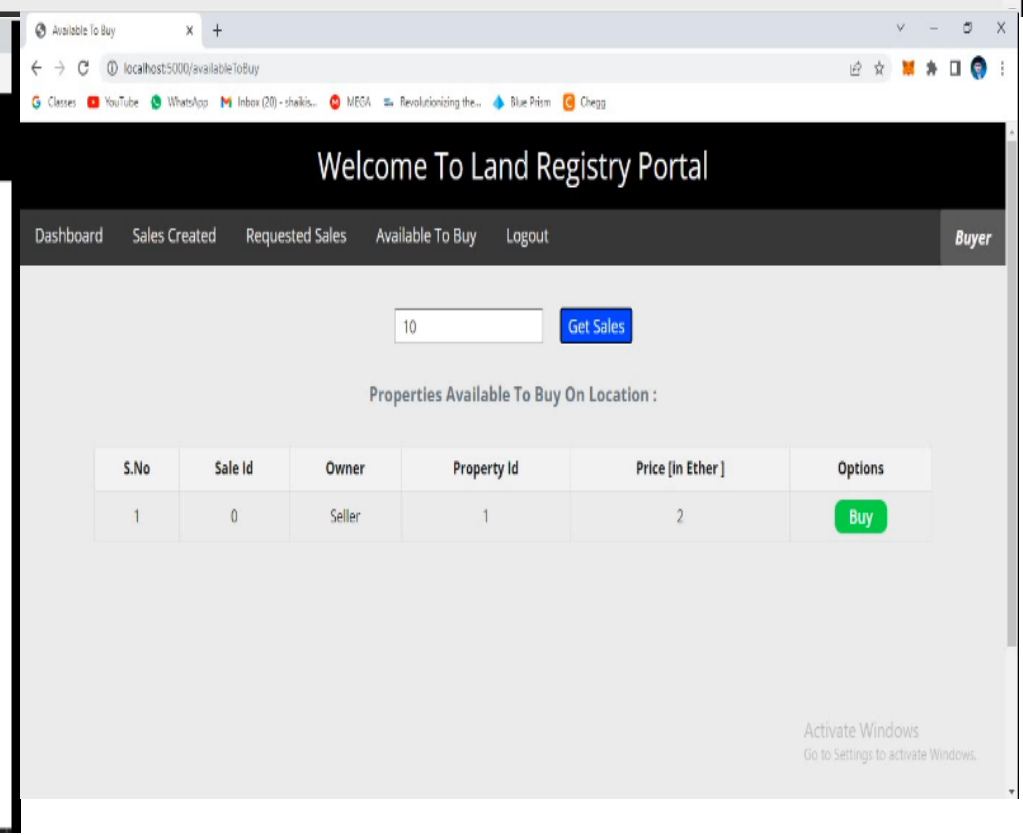
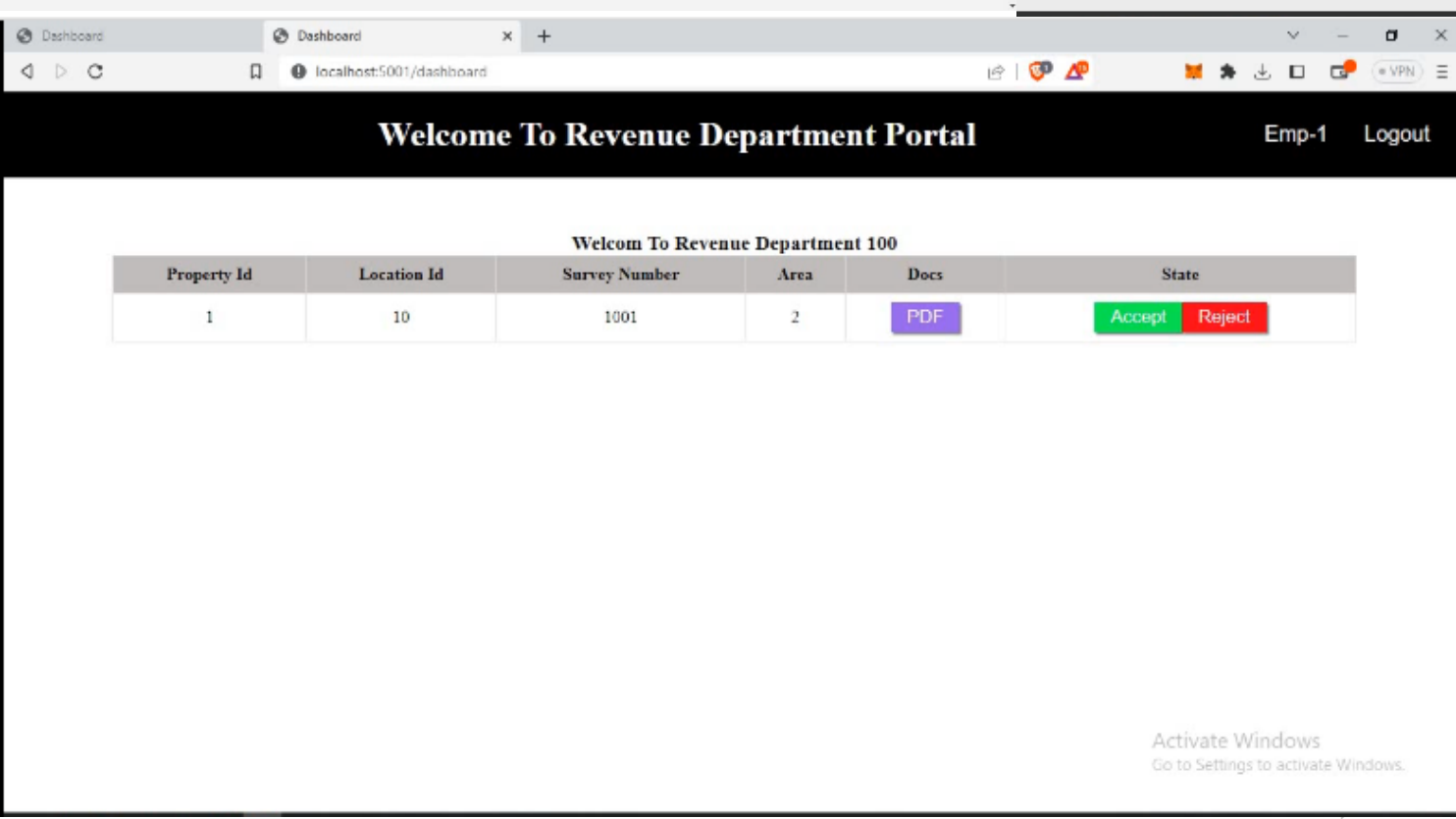
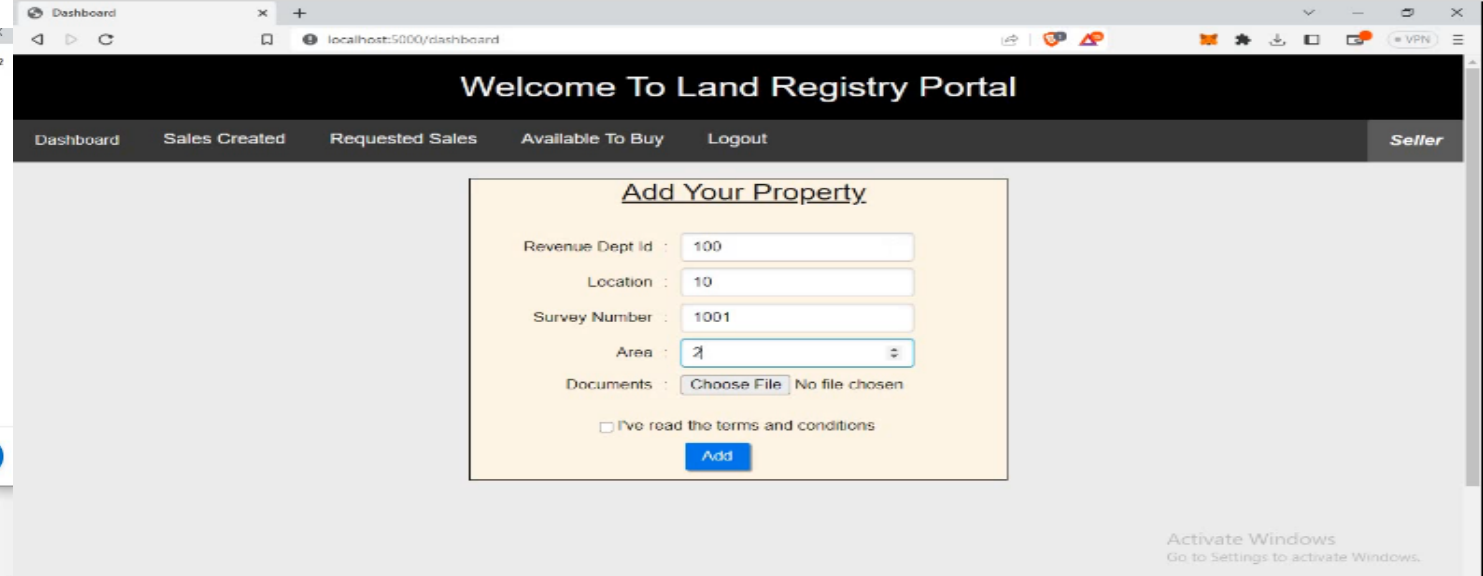
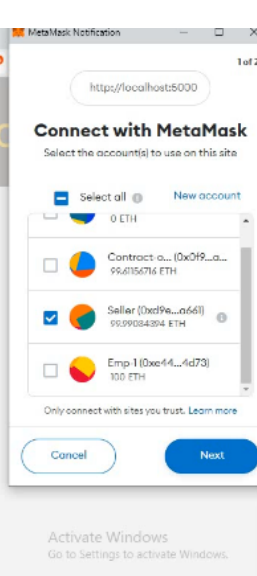
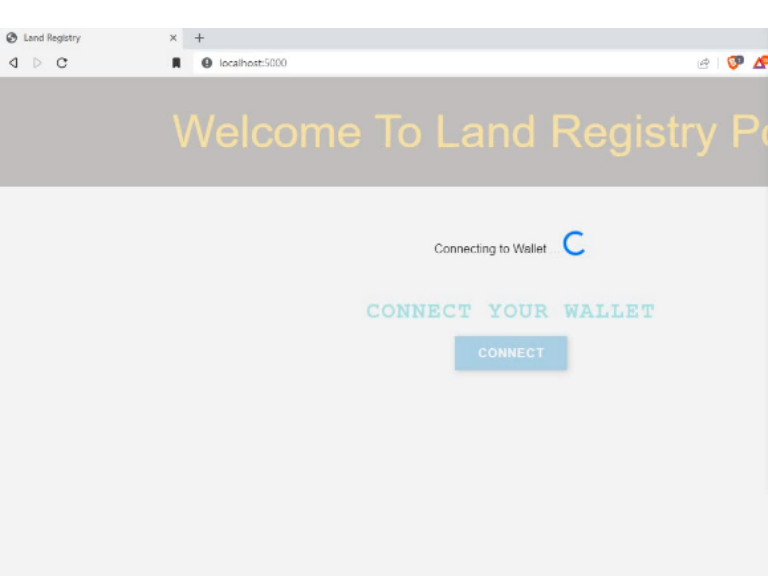
OUTCOMES OF PROPOSED :

Benefit	Description
Transparency	ML ensures fair pricing; blockchain ensures immutable records
Intelligence	Predictive analytics for land trends and disputes
Security	Fraud detection and access control via behavioral ML
Automation	Smart contracts + ML = intelligent, self-adjusting workflows
Cost & Time Reduction	Reduces manual verification and speeds up transactions



MODEL OF SYSTEM

- Frontend: User interface for buyers, sellers, and land inspectors.
- Blockchain Layer: Ethereum smart contracts for ownership and transaction history.
- IPFS: Decentralized storage for land documents.
- ML Module:
 - Fraud detection using anomaly detection algorithms
 - Document classification and OCR
 - Land price prediction using regression models
- Admin Dashboard: For land inspector verification and approval.





TECHNOLOGIES USED

Technologies

Higher Level:

- Solidity for smart contracts

Frontend:

- Javascript
- React Framework
- CSS

Backend:

- Ethereum Blockchain (Truffle Suite)
- Solidity
- Ganache



METHODOLOGY AND IMPLEMENTATION

METHODOLOGY:

- Requirements Gathering
- Blockchain Platform Selection
- Smart Contract Development
- Data Storage and Retrieval
- User Interface Development
- Testing and Quality assurance
- Deployment and Maintenance

IMPLEMENTATION:

- Blockchain Platform
 - User Interface
1. Web Application
 2. Mobile Application

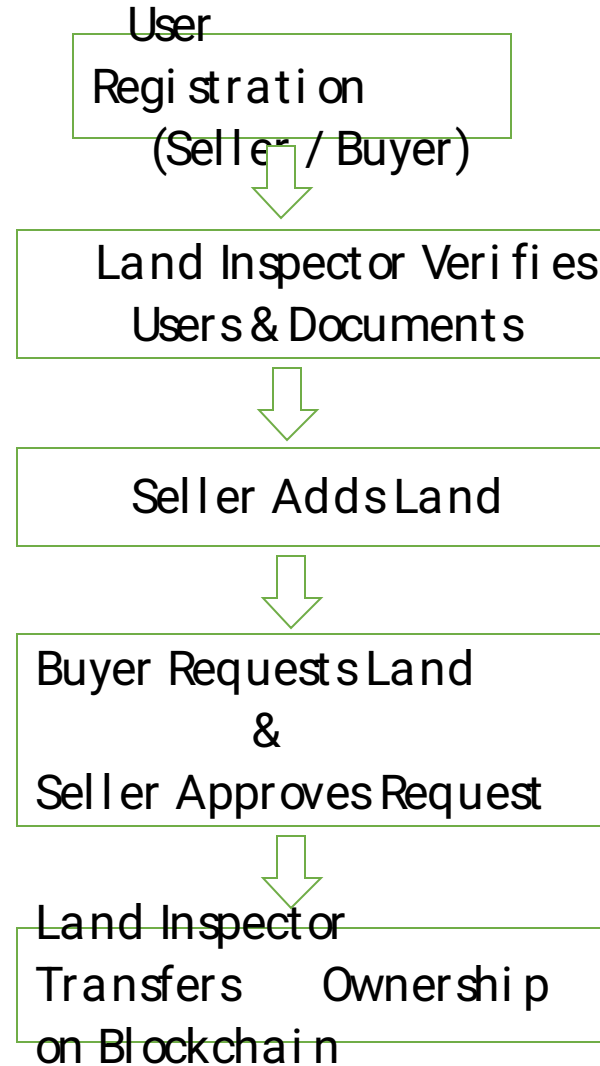


BENEFITS OF PROJECT

- Tamper-proof ownership records
- AI-powered fraud detection
- Reduced transaction time and cost
- Enhanced security and trust
- Predictive analytics for land value and disputes



APPLICATION FEATURES





SCOPE OF YOUR PRODUCT FOR NEXT

The scope of a land record management system using blockchain is vast and can have a significant impact on various stakeholders and industries

Government and Public Sector

1. Land registration
2. Property tax collection
3. Land use planning and management

Real Estate and Property Development

Property ownership and transfer

Real estate transactions

- AI-powered fraud detection
- Reduced transaction time and cost
- Enhanced security and trust
- Predictive analytics for land value and disputes



PROJECT DESCRIPTION

- This project aims to develop a secure, transparent, and intelligent land registration system by integrating blockchain technology with machine learning (ML).
- The system addresses the limitations of traditional land registries—such as fraud, inefficiency, and lack of transparency—by leveraging the immutability of blockchain and the predictive power of ML.



CONCLUSION

- Blockchain ensures secure, tamper-proof, and transparent land ownership records.
- Machine learning adds intelligence through fraud detection, document verification, and land valuation.
- The integrated system reduces manual errors, bureaucracy, and corruption.
- Real-world pilots in India, Georgia, and Sweden show promising results.
- This solution promotes trust, efficiency, and future-ready governance.
- A blockchain + ML land registry system is not just innovative—it's necessary for modern land administration.



THANK YOU